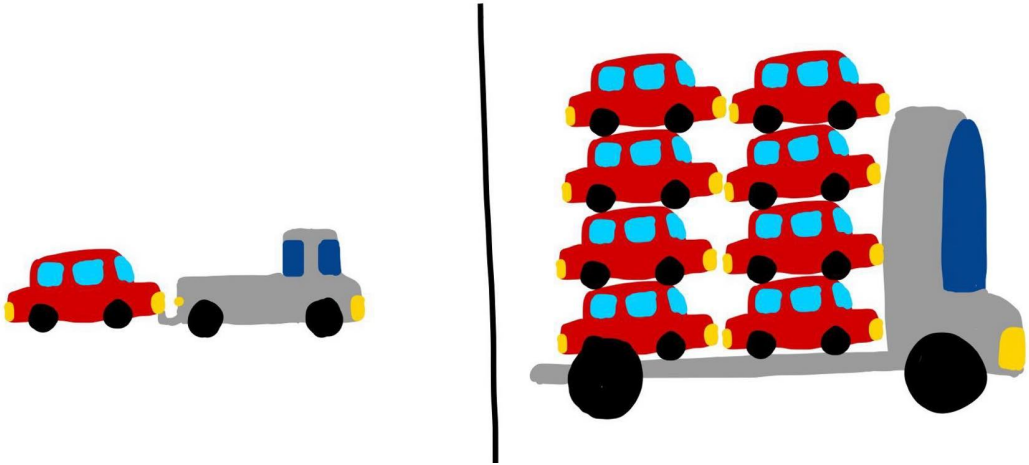
Optimizing optimizers

in PyTorch

Runtime vs memory

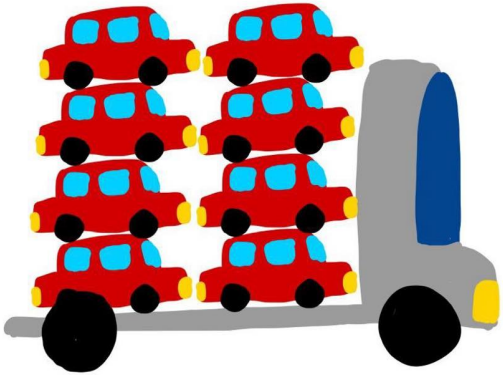
Runtime and memory usage are often at odds with each other



you're towing 512 cars from A to B. which truck do you take?

Runtime vs memory

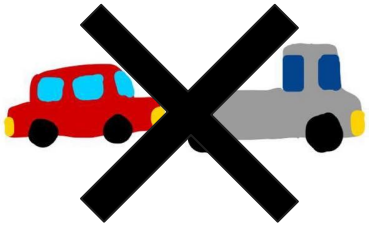
Runtime and memory usage are often at odds with each other

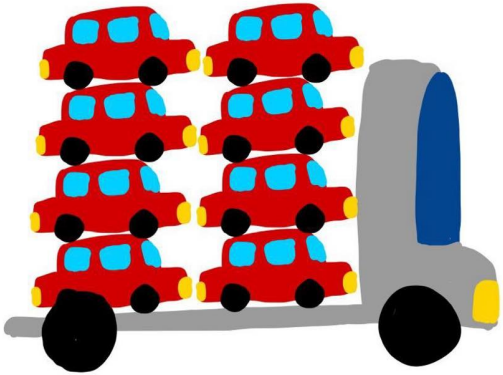


you're towing 512 cars from A to B. which truck do you take?
what if the only way to B had a low clearance bridge?

Runtime vs memory

Runtime and memory usage are often at odds with each other





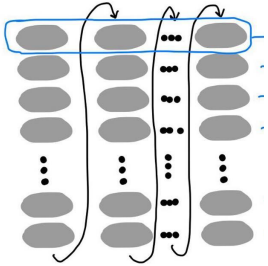
Today, we focus on speeeeeeeeed!

yes this does mean memory will take a hit, disclaimer

Fusion, the high level idea

your simplest optimizer

for loop/single tensor

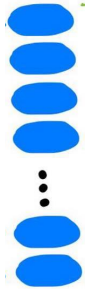


https://github.com/pytorch/pytorch/blob/b5ba80828f77c565bcda7558da97c792af32d517/torch/optim/adamw.py#L362

```
for i, param in enumerate(params):
    grad = grads[i] if not maximize else -grads[i]
    exp_avg = exp_avgs[i]
    exp_avg_sq = exp_avg_sqs[i]
    step_t = state_steps[i]

    # update step
    step_t += 1

    # Perform stepweight decay
    param.mul_(1 - lr * weight_decay)

    # Decay the first and second moment running average coefficient
    exp_avg.lerp_(grad, 1 - beta1)
    exp_avg_sq.mul_(beta2).addcmul_(grad, grad, value=1 - beta2)
    step = _get_value(step_t)

    bias_correction1 = 1 - beta1 ** step
    bias_correction2 = 1 - beta2 ** step

    step_size = lr / bias_correction1

    bias_correction2_sqrt = _dispatch_sqrt(bias_correction2)
    denom = (exp_avg_sq.sqrt() / bias_correction2_sqrt).add_(eps)

    param.addcdiv_(exp_avg, denom, value=-step_size)
```

horizontally fused optimizer

foreach



https://github.com/pytorch/pytorch/blob/b5ba80828f77c565bcda7558da97c792af32d517/torch/optim/adamw.py#L480

```
torch._foreach_add_(device_state_steps, 1)

# Perform stepweight decay
if weight_decay != 0:
    torch._foreach_mul_(device_params, 1 - lr * weight_decay)

# Decay the first and second moment running average coefficient
torch._foreach_lerp_(device_exp_avgs, device_grads, 1 - beta1)

torch._foreach_mul_(device_exp_avg_sqs, beta2)
torch._foreach_addcmul_(device_exp_avg_sqs, device_grads,
device_grads, 1 - beta2)

. . .

torch._foreach_div_(exp_avg_sq_sqrt, bias_correction2_sqrt)
torch._foreach_add_(exp_avg_sq_sqrt, eps)
torch._foreach_addcdiv_(device_params, device_exp_avgs,
exp_avg_sq_sqrt, step_size)
```

entirely fused optimizer

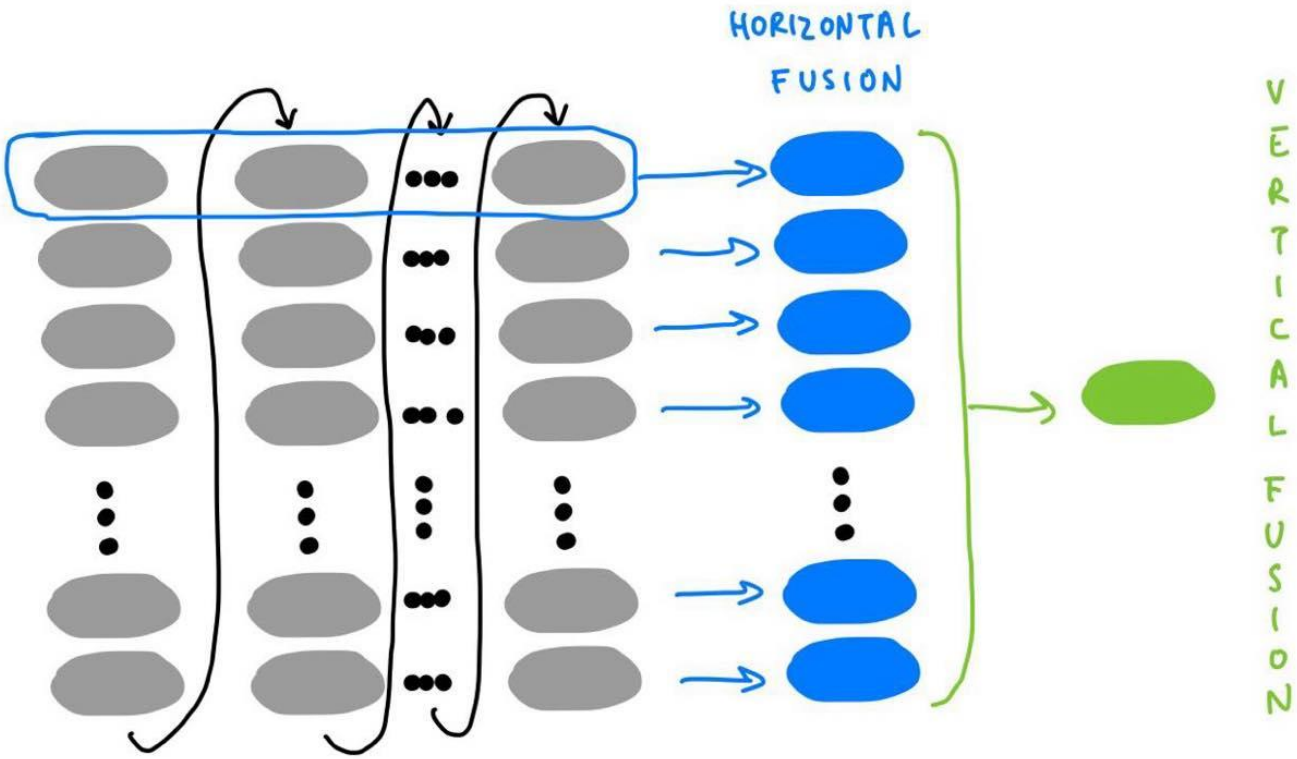
...fused

(thanks NVIDIA)

```
torch._fused_adamw_(
    device_params,
    device_grads,
    device_exp_avgs,
    device_exp_avg_sqs,
    device_max_exp_avg_sqs,
    device_state_steps,
    amsgrad=amsgrad,
    lr=lr,
    beta1=beta1,
    beta2=beta2,
    weight_decay=weight_decay,
    eps=eps,
    maximize=maximize,
    grad_scale=device_grad_scale,
    found_inf=device_found_inf,
)
```

https://github.com/pytorch/pytorch/blob/b5ba80828f77c565bcda7558da97c792af32d517/torch/optim/adamw.py#L616

The Gist: the fewer kernels you launch on CUDA, the faster.

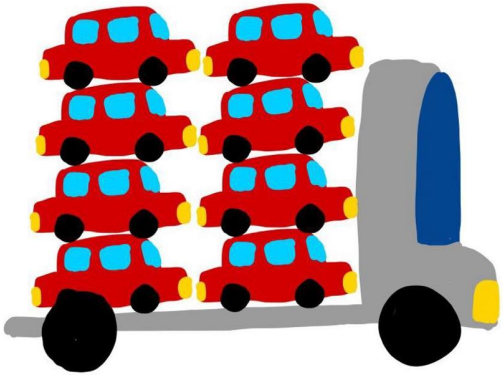


Fusion, the nitty gritty

starting with multi_tensor_apply

you know how mitochondria is the powerhouse of the cell?
multi_tensor_apply is the powertruck of our speedy optimizers.

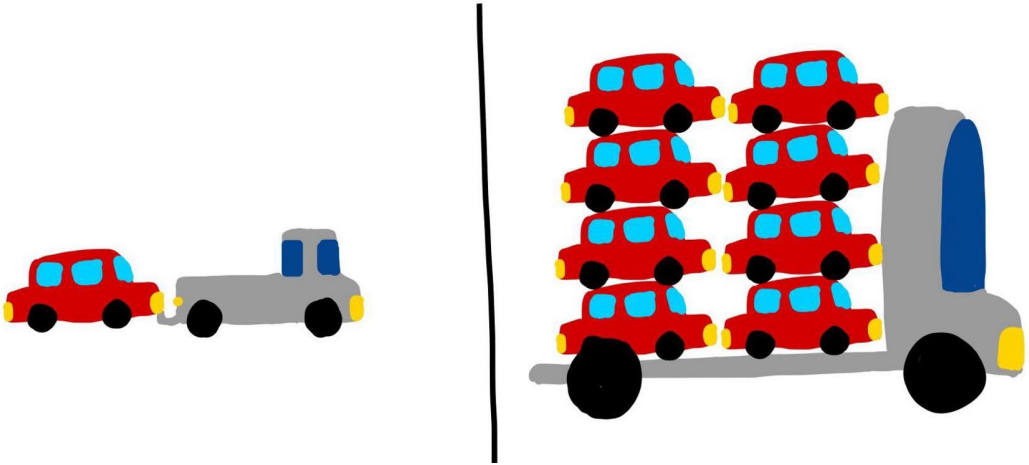
CPU



GPU

multi_tensor_apply allows us to operate over a list of Tensors vs a single tensor.

Example with torch.add



```
add(
    Tensor self, Tensor other, *,
    Scalar alpha=1
) -> Tensor
```

```
_foreach_add(
    Tensor[] self, Tensor[] other, *,
    Scalar alpha=1
) -> Tensor[]
```

Under the hood in CUDA

```
add(Tensor self, Tensor other, *,
Scalar alpha=1) -> Tensor
```

A simplified CUDA kernel signature, assuming we have float Tensors:

```
__device__ void add_kernel(
        float* self,
        float* other,
        float* res,
        float alpha=1) {
...
}
```

```
_foreach_add(Tensor[] self,
Tensor[] other, *, Scalar alpha=1)
-> Tensor[]
```

How would you write this one?

Attempt 1: use std::vector

```
_foreach_add(Tensor[] self, Tensor[] other, *, Scalar alpha=1) -> Tensor[]

__device__ void _foreach_add_kernel(
        std::vector<float*> self,
        std::vector<float*> other,
        std::vector<float*> res,
        float alpha=1) {
...
}
```

Does this work?

```
add(Tensor self, Tensor
other, *, Scalar alpha=1) ->
Tensor

__device__ void add_kernel(
        float* self,
        float* other,
        float* res,
        float alpha=1) {
...
}
```

Attempt 1: use std::vector

```
_foreach_add(Tensor[] self, Tensor[] other, *, Scalar alpha=1) -> Tensor[]

__device__ void _foreach_add_kernel(
        std::vector<float*> self,
        std::vector<float*> other,
        std::vector<float*> res,
        float alpha=1) {
...
}
```

```
add(Tensor self, Tensor
other, *, Scalar alpha=1) ->
Tensor

__device__ void add_kernel(
        float* self,
        float* other,
        float* res,
        float alpha=1) {
...
```

Does this work?
No because CUDA doesn't recognize std::vectors!

Attempt 2: fine i'll use a C-style array then

```
_foreach_add(Tensor[] self, Tensor[] other, *, Scalar alpha=1) -> Tensor[]

__device__ void _foreach_add_kernel(
        float** self,
        float** other,
        float** res,
        float alpha=1) {
...
}
```

Does this work?

```
add(Tensor self, Tensor
other, *, Scalar alpha=1) ->
Tensor

__device__ void add_kernel(
        float* self,
        float* other,
        float* res,
        float alpha=1) {
...
}
```

Attempt 2: fine i'll use a C-style array then

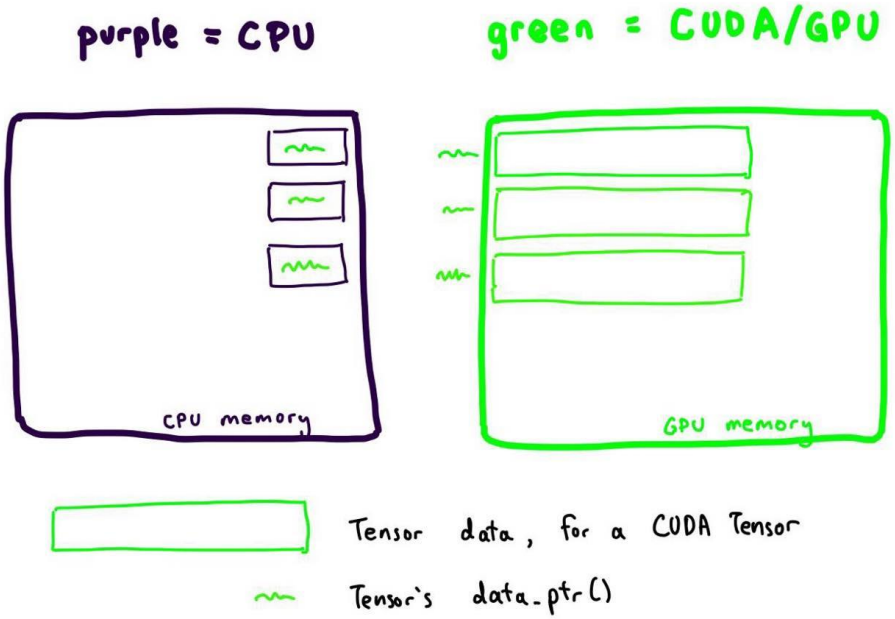
```
_foreach_add(Tensor[] self, Tensor[] other, *, Scalar alpha=1) -> Tensor[]

__device__ void _foreach_add_kernel(
        float** self,
        float** other,
        float** res,
        float alpha=1) {
...
}
```

```
add(Tensor self, Tensor
other, *, Scalar alpha=1) ->
Tensor

__device__ void add_kernel(
        float* self,
        float* other,
        float* res,
        float alpha=1) {
...
}
```

Does this work?
Nope! This will cause an Illegal Memory Access (IMA) because the outer pointer * is a CPU address!

Attempt 2: fine i'll use a C-style array then

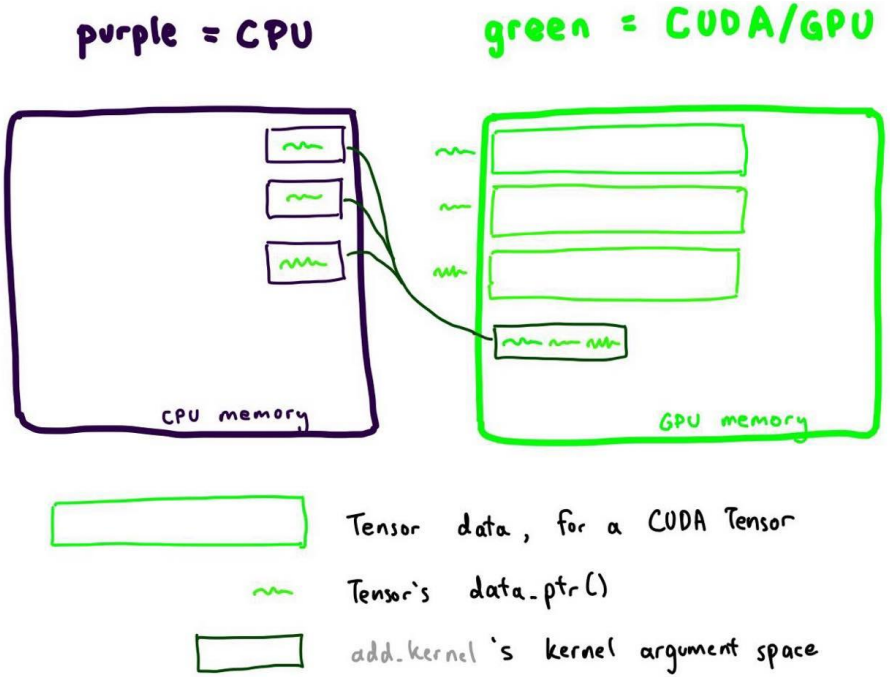
Look again at the add_kernel.




```
add(Tensor self, Tensor
other, *, Scalar alpha=1) ->
Tensor

__device__ void add_kernel(
        float* self,
        float* other,
        float* res,
        float alpha=1) {
...
```

Attempt 2: fine i'll use a C-style array then

When we dereference in the CUDA kernel, it is OK! The address is in GPU.

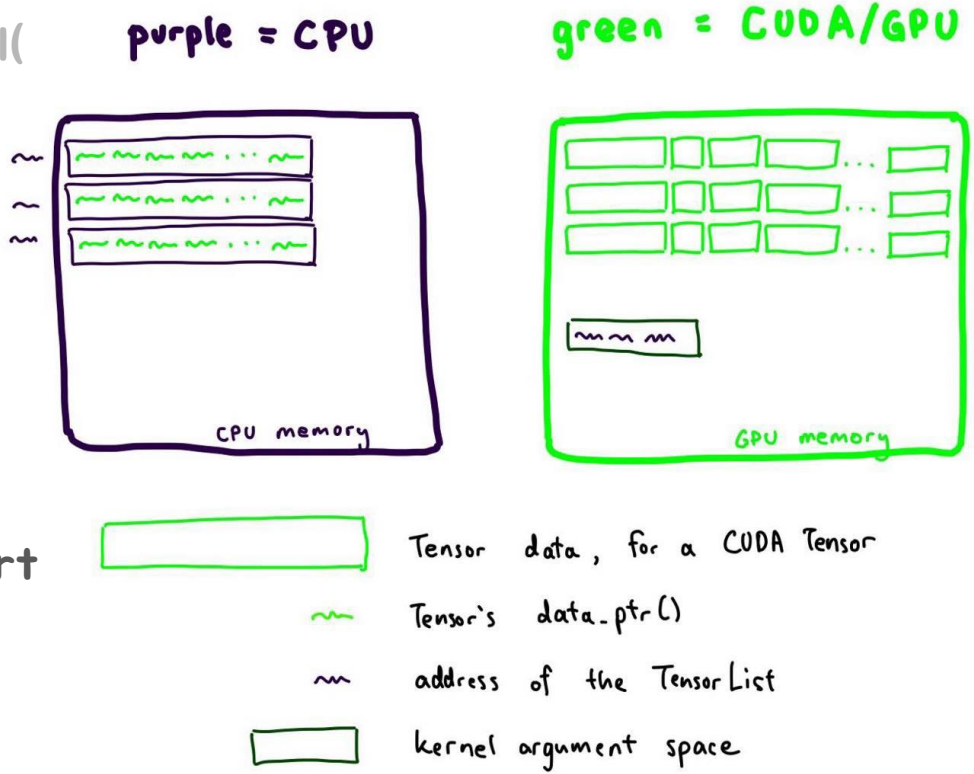



```
add(Tensor self, Tensor
other, *, Scalar alpha=1) ->
Tensor

__device__ void add_kernel(
        float* self,
        float* other,
        float* res,
        float alpha=1) {
...
```

you're going to dereference in here!

Attempt 2: fine i'll use a C-style array then

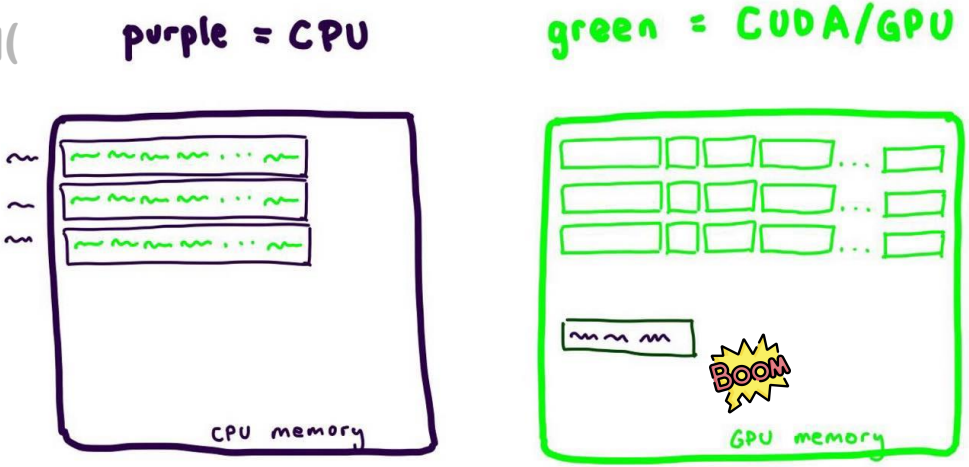
```
__device__ void _foreach_add_kernel(
        float** self,
        float** other,
        float** res,
        float alpha=1) {
...
}
```



But not so our _foreach_add kernel. While Tensors live in GPU, the list part of TensorLists live in CPU.

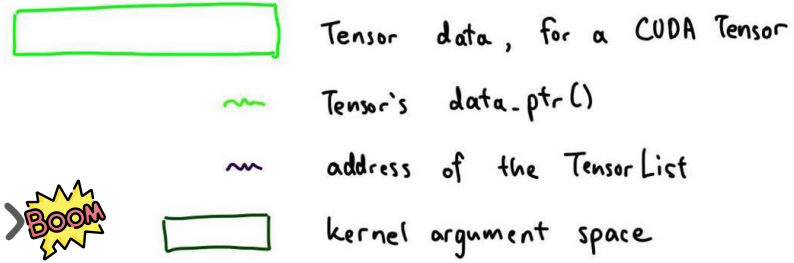
Attempt 2: fine i'll use a C-style array then

```
__device__ void _foreach_add_kernel(
        float** self,
        float** other,
        float** res,
        float alpha=1) {
... you're going to dereference in here!
}
```



But not so our _foreach_add kernel.
While Tensors live in GPU, the list part of TensorLists live in CPU.

Dereferencing a CPU address from GPU =>



Attempt 3: pass by chonky boi (not reference)

```
struct TensorListMetadata {
  const float* addresses[3][NUM_TENSORS];
};

<add all the addresses into the struct>

__device__ void _foreach_add_kernel(
          TensorListMetadata tlm,
          float alpha=1) {
...
}
```

Does this work?

```
add(Tensor self, Tensor
other, *, Scalar alpha=1) ->
Tensor

__device__ void add_kernel(
          float* self,
          float* other,
          float* res,
          float alpha=1) {
...
```

Attempt 3: pass by chonky boi (not reference)

```
struct TensorListMetadata {
  const float* addresses[3][NUM_TENSORS];
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__device__ void _foreach_add_kernel(
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...
}
```
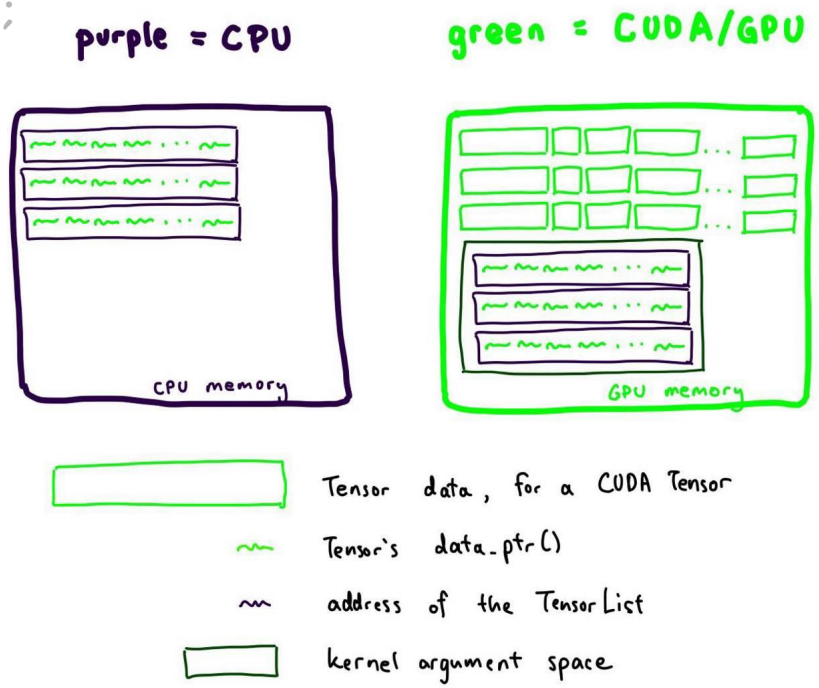



Does this work? It passes CI! Yay!

~ the end ~

~ the end ~

jk

I actually did land a PR like this and it got reverted :(

Cuz some! how! an illegal memory access happened

for some models (like timm_efficientdet).

I minified the repro to the following and played around with N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

Now would you join me on my (binary) search!

(yes, this example is for norm and not add, but the principle is the same!)

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌
N = 450 💥

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌
N = 450 💥
N = 425 💥

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌
N = 450 💥
N = 425 💥
N = 412 👌

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌
N = 450 💥
N = 425 💥
N = 412 👌
N = 420 👌

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌
N = 450 💥
N = 425 💥
N = 412 👌
N = 420 👌
N = 423 👌

Let's binary search over N.

```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌
N = 450 💥
N = 425 💥
N = 412 👌
N = 420 👌
N = 423 👌
N = 424 💥

Let's binary search over N.

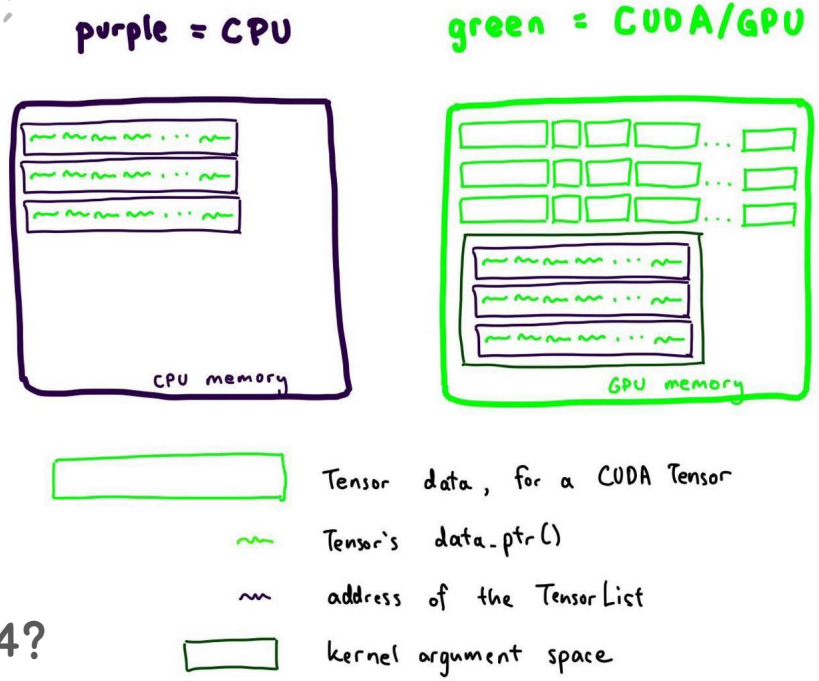
```
params = [torch.rand(2, 3, device="cuda") for _ in range(N)]
torch._foreach_norm(params, ord=1)
torch.cuda.synchronize()
```

N = 500 💥
N = 256 👌
N = 400 👌
N = 450 💥
N = 425 💥
N = 412 👌
N = 420 👌
N = 423 👌
N = 424 💥

what is so special about these numbers?
what could be going on here?

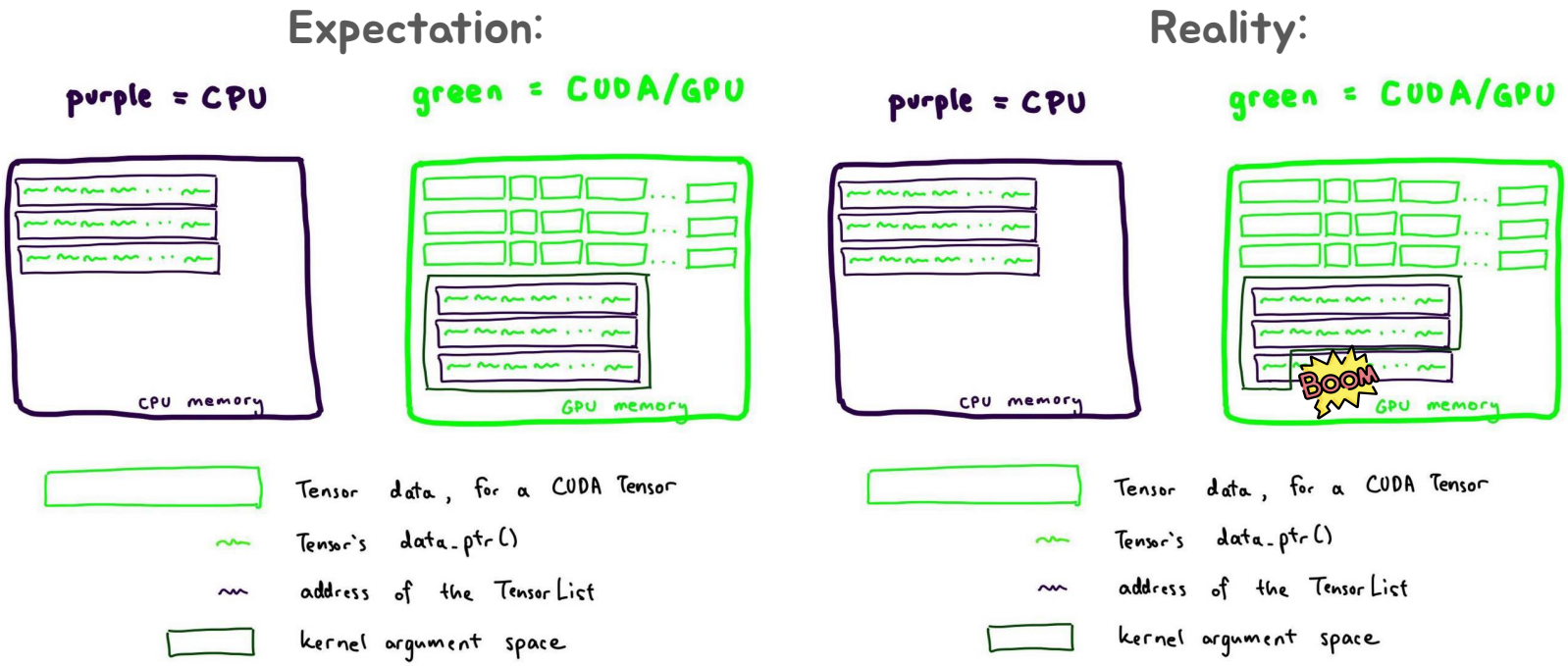
Attempt 3: pass by chonky boi (not reference) cont.

```
struct TensorListMetadata {
  const float* addresses[3][NUM_TENSORS];
};

<add all the addresses into the struct>

__device__ void _foreach_add_kernel(
          TensorListMetadata tlm,
          float alpha=1) {
...
}
```



Does this work? Only if NUM_TENSORS < 424?

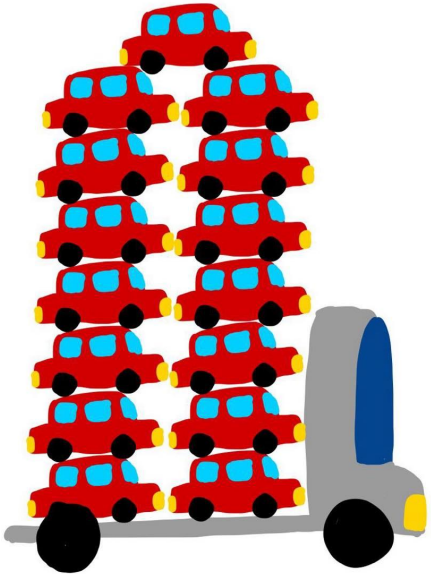
Attempt 3: pass by chonky boi (not reference) cont.



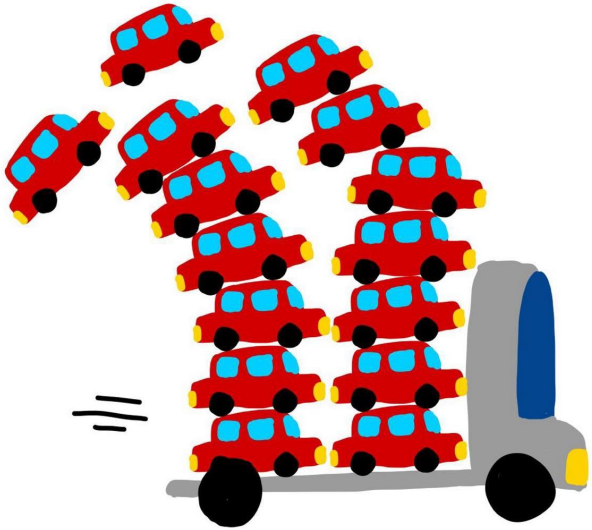
Fun fact: CUDA Kernel argument space has a max limit of 4KB 🥲

Attempt 3: pass by chonky boi (not reference) cont.

Expectation:

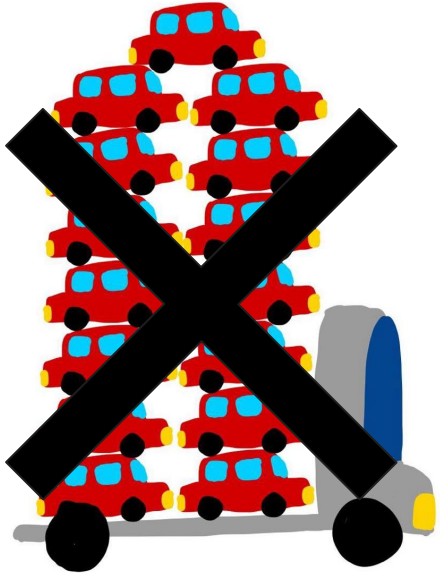


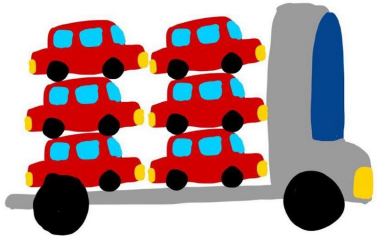
Reality:

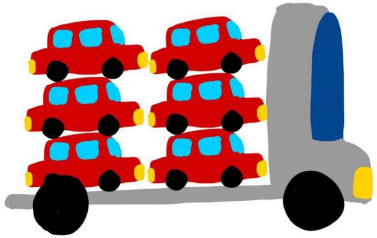


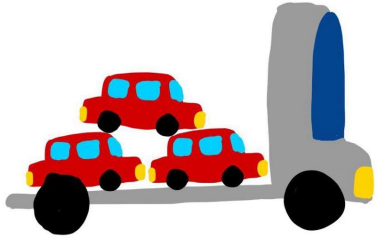
Fun fact: CUDA Kernel argument space has a max limit of 4KB 🥲 so what now?

Attempt 4: just launch more kernels; make more trips





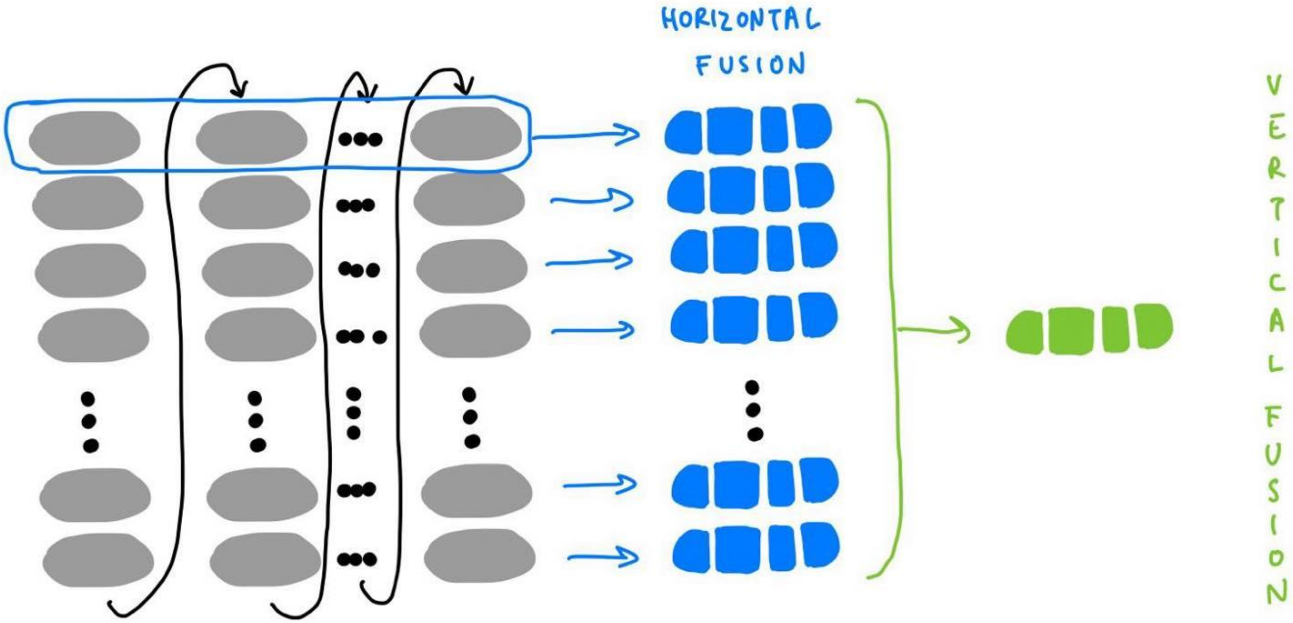




```
struct TensorListMetadata {
  const float*
addresses[3][MAX_NUM_TENSORS];
};

<make multiple structs>
<add a chunk of the addresses to each>
<launch the kernel multiple times>

__device__ void _foreach_add_kernel(
          TensorListMetadata tlm,
          float alpha=1) {

...
}
```

Attempt 4 *is* what we do today. But we could do better.

While we claim to horizontally fuse into 1 kernel...we often end up with more:

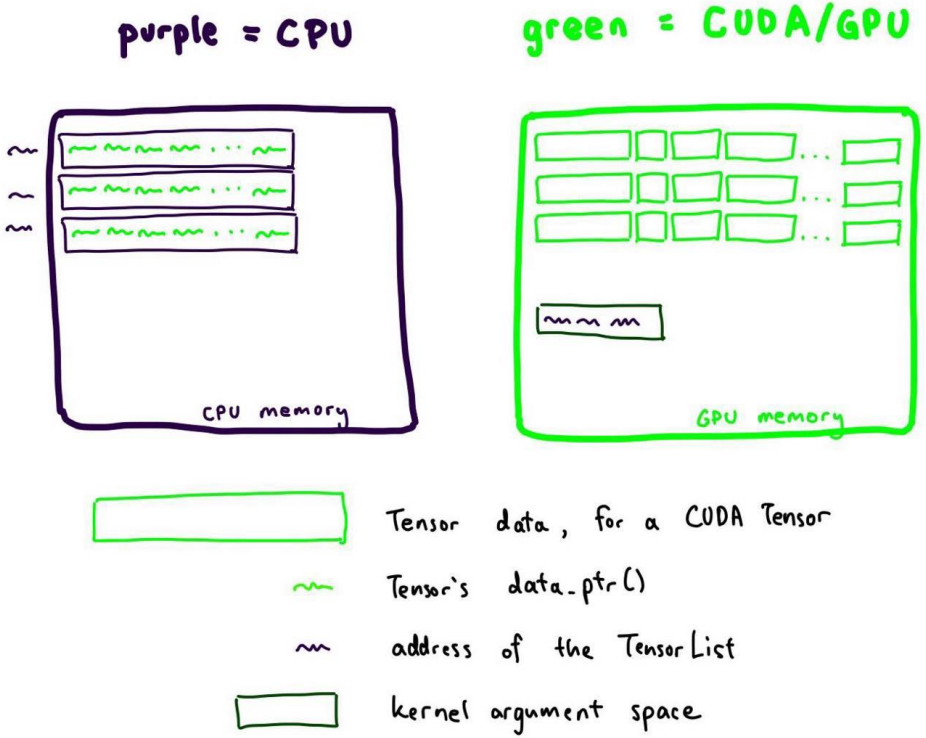


How: revisit attempt 2

```
__device__ void _foreach_add_kernel(
        float** self,
        float** other,
        float** res,
        float alpha=1) {
...
}
```

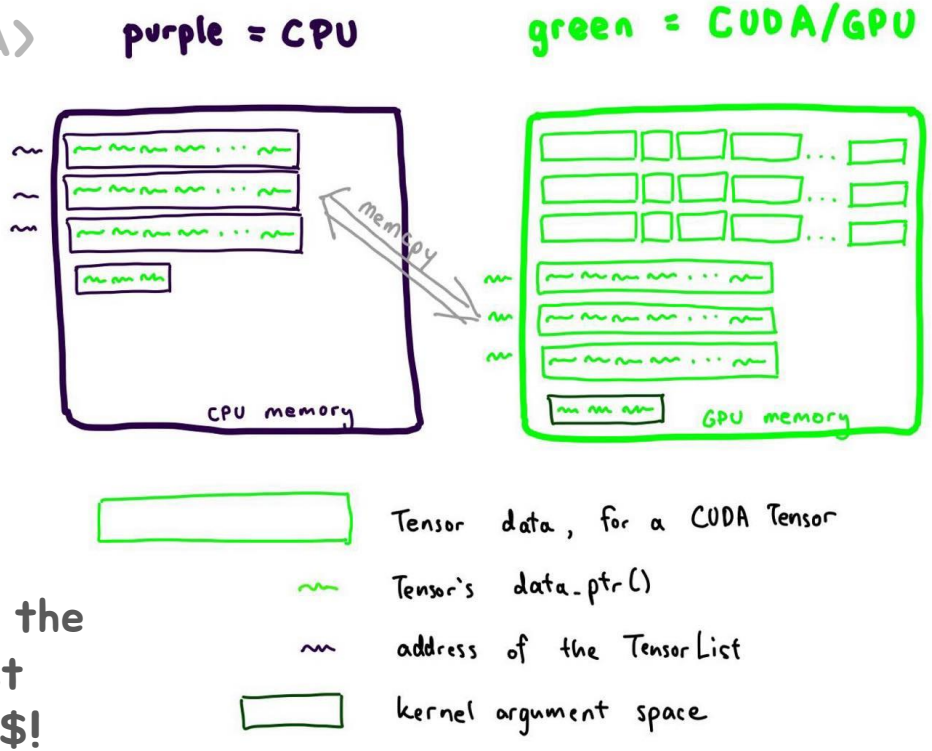
We wanna turn those purple * to green *.

How? Move them to CUDA beforehand!
(thanks Yifu Wang!)



How: revisit attempt 2

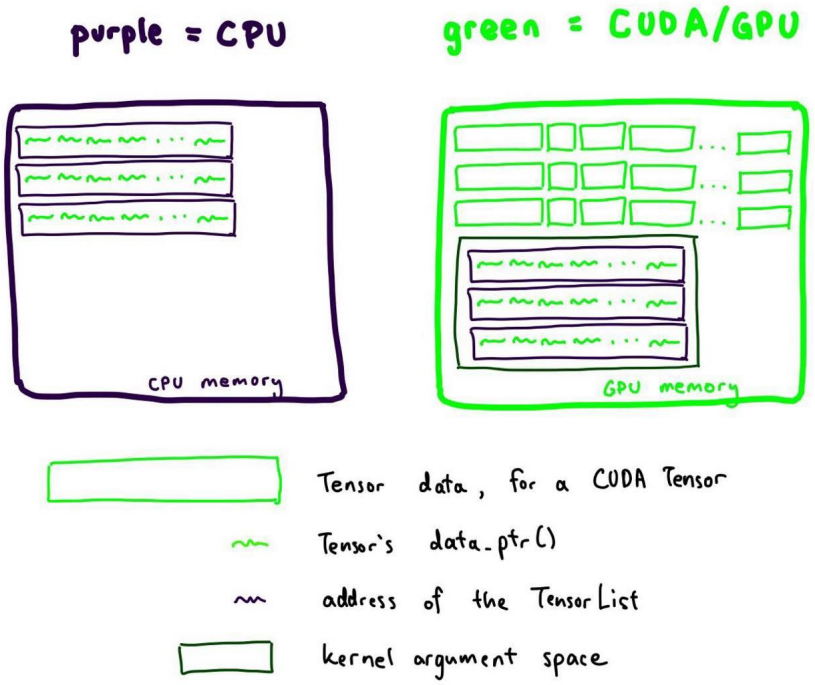
```
<memcpy the lists of addresses to CUDA>

__device__ void _foreach_add_kernel(
        float** self,
        float** other,
        float** res,
        float alpha=1) {
...
}
```



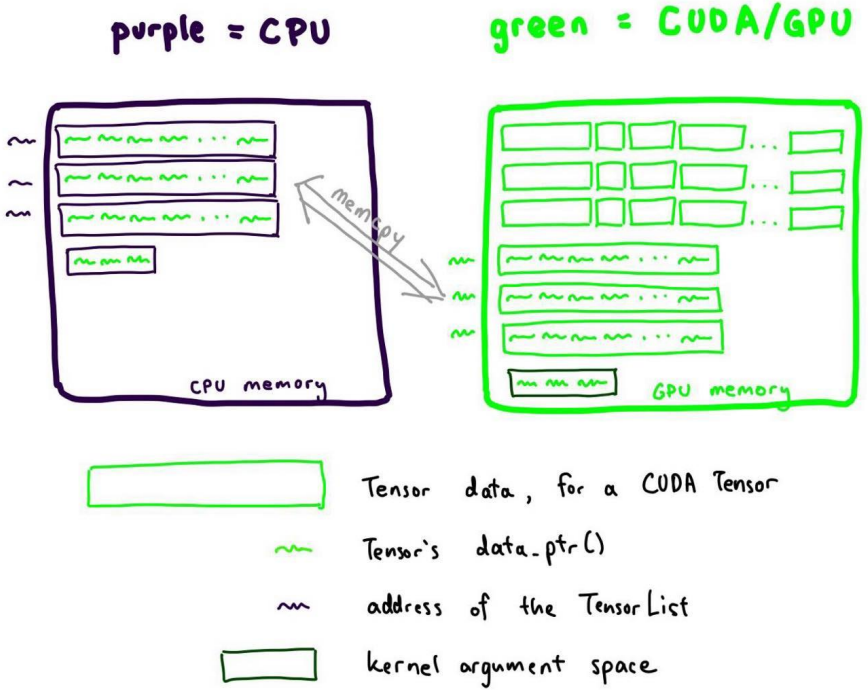
We thereby avoid the 4KB constraint in the kernel argument to enable launching just one kernel. But remember, memcpy is $$$!

Conclusion: we will be doing a mix of struct + memcpy

if it fits, use the struct

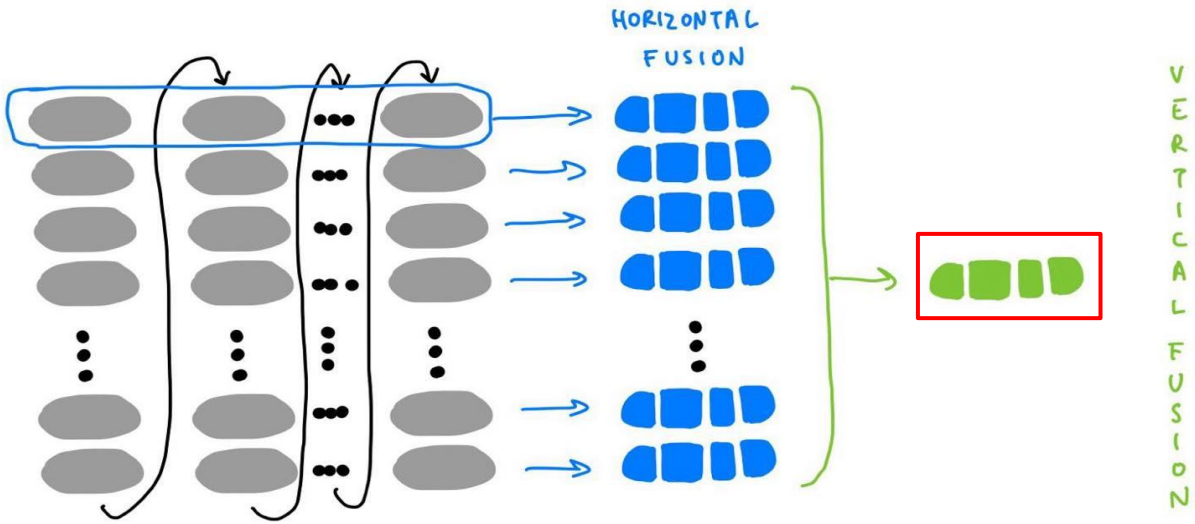


otherwise, take the memcpy hit



Did you notice I split up the fused implementation too?

This is cuz our fastest fused impls also rely on multi_tensor_apply!



Did you notice I split up the fused implementation too?

Whereas _foreach_add will call multi_tensor_apply with a Callable that does addition, _fused_adamw will call multi_tensor_apply with a bigger Callable.

```
multi_tensor_apply_kernel<<<
    loc_block_info,
    kBlockSize,
    0,
    at::cuda::getCurrentCUDAStream()>>>(tensorListMeta, callable, args...);
C10_CUDA_KERNEL_LAUNCH_CHECK();
```

Did you notice I split up the fused implementation too?

Whereas _foreach_add will call multi_tensor_apply with a Callable that does addition, _fused_adamw will call multi_tensor_apply with a bigger Callable.

```
AT_DISPATCH_FLOATING_TYPES_AND2(
    kHalf,
    kBFloat16,
    params[0].scalar_type(),
    "fused_adamw_kernel_cuda",
    [&]() {
      multi_tensor_apply_for_fused_optimizer<4>(
          tensor_lists,
          state_steps,
          FusedAdamMathFunctor<scalar_t, 4, ADAM_MODE::ADAMW, false>(),
          lr_ptr,
          1.0, // unused
          beta1,
          beta2,
          weight_decay,
          eps,
          maximize,      You, 6 months ago • [optim] FusedAdam/W accepts
          grad_scale_ptr,
          found_inf_ptr);
    });
```

so let's peek at FusedAdamMathFunctor

```cpp
struct FusedAdamMathFunctor {
  C10_DEVICE __forceinline__ void operator()(
    const float* found_inf_ptr) {
    const auto tensor_loc = tl.block_to_tensor[blockIdx.x];
    const auto chunk_idx = tl.block_to_chunk[blockIdx.x];
    const double lr_double = lr_ptr ? *lr_ptr : lr;

    if (found_inf_ptr && *found_inf_ptr == 1) {
      return;
    }
    const auto [bias_correction1: <dependent type>, bias_correction2_sqrt: <dependent type>] =
        [&]() -> std::pair<double, double> {
      auto* step_count: const float * =
          reinterpret_cast<const float*>(tl.state_steps_addresses[tensor_loc]);        Masaki Kozuki, 2
      const auto bias_correction1: double const = 1 - at::native::pow_(base: beta1, exp: *step_count);
      const auto bias_correction2: double const = 1 - at::native::pow_(base: beta2, exp: *step_count);
      const auto bias_correction2_sqrt: const double = std::sqrt(x: bias_correction2);
      return {a: bias_correction1, b: bias_correction2_sqrt};
    }();

    scalar_type* args[depth];
    scalar_type r_args[depth][kILP];
    const auto n = tl.numel_for_tensor[tensor_loc] - chunk_idx * chunk_size;

    const bool all_aligned{
        init_args<depth>(args, tl, chunk_idx, chunk_size, tensor_loc)};
    if ((n % kILP == 0) && (chunk_size % kILP == 0) && all_aligned) {
      for (int64_t i_start = threadIdx.x;
          i_start * kILP < n && i_start * kILP < chunk_size;
          i_start += blockDim.x) {
#pragma unroll
        for (int i = 0; i < depth; i++) {
          load_store(dst: r_args[i], src: args[i], dst_offset: 0, src_offset: i_start);
        }
        adam_math<scalar_type, opmath_t, depth, adam_mode, amsgrad>(
            r_args,
            lr: lr_double,
```

```cpp
C10_DEVICE inline void adam_math(
#pragma unroll
  for (int ii = 0; ii < kILP; ii++) {
    // Load values.
    opmath_t param = static_cast<opmath_t>(r_args[kParamIdx][ii]);
    opmath_t grad = static_cast<opmath_t>(r_args[kGradIdx][ii]);
    if (grad_scale_ptr) {
      grad /= (static_cast<double>(*grad_scale_ptr));
    }
    const opmath_t grad_to_store = grad;
    if (maximize) {
      grad = -grad;
    }
    opmath_t exp_avg = static_cast<opmath_t>(r_args[kExpAvgIdx][ii]);
    opmath_t exp_avg_sq = static_cast<opmath_t>(r_args[kExpAvgSqIdx][ii]);
    opmath_t max_exp_avg_sq;
    if (amsgrad) {
      max_exp_avg_sq = static_cast<opmath_t>(r_args[kMaxExpAvgSqIdx][ii]);
    }
    // Update param, grad, 1st and 2nd order momentum.
    if (weight_decay != 0) {
      if constexpr (adam_mode == ADAM_MODE::ORIGINAL) {
        grad += param * weight_decay;
      } else if constexpr (adam_mode == ADAM_MODE::ADAMW) {
        param -= lr * weight_decay * param;
      }
    }
    // todo(crcrpar): use lerp
    // ref: https://developer.nvidia.com/blog/lerp-faster-cuda/
    exp_avg = beta1 * exp_avg + (1 - beta1) * grad;
    exp_avg_sq = beta2 * exp_avg_sq + (1 - beta2) * grad * grad;        Masak
    const opmath_t step_size = lr / bias_correction1;
    opmath_t denom;
    if (amsgrad) {
      max_exp_avg_sq = std::max(max_exp_avg_sq, exp_avg_sq);
      denom = (std::sqrt(max_exp_avg_sq) / bias_correction2_sqrt) + eps;
```

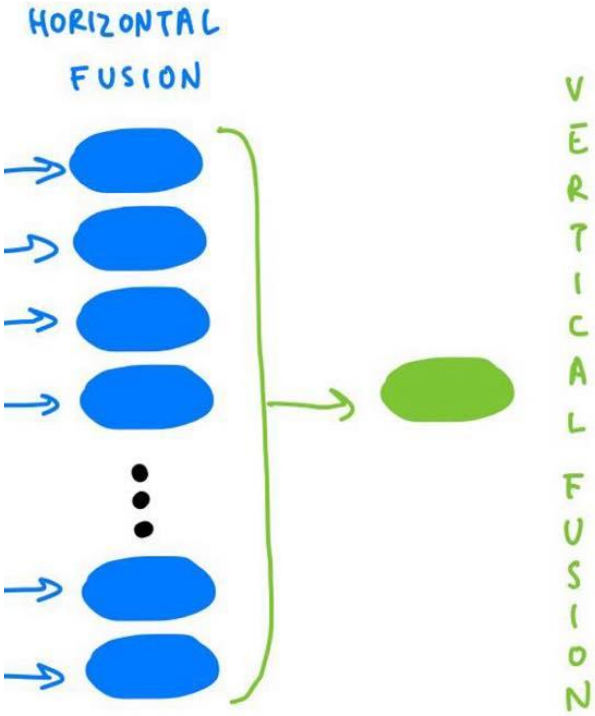
...that was very manual.

What if we could automate vertical fusion?

with just 1 line?

Enter torch.compile()

torch.compile()'s strength is vertical fusion



How do I use torch.compile() with optimizers?

```
optimizer = torch.optim.AdamW(params)

@torch.compile(fullgraph=False)
def compiled_step():
    optimizer.step()
```

Now call compiled_step instead of optimizer.step() in your training loop. That's it!

(okay I suppose that was 2 lines)

Inductor will generate a large triton kernel

```
xpid_offset = xpid - 0
xnumel = 512
xoffset = xpid_offset * XBLOCK
xindex = xoffset + tl.arange(0, XBLOCK)[:]
xmask = xindex < xnumel
x0 = xindex
tmp0 = tl.load(in_ptr0 + (x0), xmask)
tmp1 = tl.load(in_ptr1 + (x0), xmask)
tmp6 = tl.load(in_ptr2 + (x0), xmask)
tmp13 = tl.load(in_ptr3 + (x0), xmask)
tmp17 = tl.load(in_ptr4 + (0))
tmp18 = tl.broadcast_to(tmp17, [XBLOCK])
tmp2 = tmp1 - tmp0
tmp3 = 0.09999999999999998
tmp4 = tmp2 * tmp3
tmp5 = tmp0 + tmp4
tmp7 = 0.999
tmp8 = tmp6 * tmp7
tmp9 = tmp1 * tmp1
tmp10 = 0.0010000000000000009
tmp11 = tmp9 * tmp10
tmp12 = tmp8 + tmp11
tmp14 = 0.99999
tmp15 = tmp13 * tmp14
tmp16 = tl.sqrt(tmp12)
tmp19 = tl.math.pow(tmp7, tmp18)
tmp20 = 1.0
tmp21 = tmp19 - tmp20
tmp22 = -tmp21
tmp23 = tl.sqrt(tmp22)
tmp24 = tmp16 / tmp23
tmp25 = 1e-08
tmp26 = tmp24 + tmp25
tmp27 = 0.9
tmp28 = tl.math.pow(tmp27, tmp18)
tmp29 = tmp28 - tmp20
tmp30 = 0.001
tmp31 = tmp29 / tmp30
tmp32 = 1 / tmp31
tmp33 = tmp26 / tmp32
tmp34 = tmp5 / tmp33
tmp35 = tmp15 + tmp34
tl.store(out_ptr0 + (x0), tmp5, xmask)
tl.store(out_ptr3 + (x0), tmp35, xmask)
tl.store(out_ptr4 + (x0), tmp12, xmask)
```

When does it work? (or not work?)

- You must have CUDA capability 7.0+ for Triton
- All optimizers in pytorch/pytorch with a foreach implementation are now compilable
  - So everything except L-BFGS and SparseAdam
- Vertical fusion of *any* sequence of supported _foreach_* ops should work!
  - try out your experimental optimizers!
  - open an issue when this isn't true

Compiled optimizers is in beta! Try it out and complain lots here!

So should you stop learning CUDA?

So should you stop learning CUDA?

NO! NO! NO!

thanks! questions?